

DIGITAL COMPRESSION COLUMN LOAD CELL FOR WEIGHBRIDGES / TRUCK SCALES

capacities 30t, 40t



This **digital** stainless steel self-centering rocker column load cell is fully welded and hermetically sealed, with ingress protection to IP68 and IP69K standards. It is an ideal solution for weighbridges and truck scales, thanks to its OIML R60 Class C4 approval (4000 divisions) and optional Class C5 approval (5000 divisions). The load cell features built-in lightning surge arrestors and a durable polyurethane cable. Its **open protocol** allows it to be used with almost any weight indicator. Anti-rotation is ensured by two flat faces on either side of the load cell base, which aligns with corresponding flat faces in the lower mounting cup.

Digital load cells offer several advantages over analogue load cells, including faster and simpler corner adjustments and the ability to diagnose individual load cell faults.

For use in weighbridges/truck scales a set of nickel-plated steel mounting adapters, centering discs and bolts are available as an optional item.

In aggressive environments where stress corrosion or acid attack could be a concern, a special Parylene coating can be specified as an option.

- Open digital protocol
- Fully welded and hermetically sealed to IP68 and IP69K
- In-built surge arrestors for lightning protection
- Simple and fast installation
- Configuration and software can be updated using a serial interface
- Self-centering rocker column design with anti-rotation faces
- Approval to OIML R60 C4 (4000 divisions) optional C5 (5000 divisions)
- Mounting cup accessories available
- Digital interface RS485 (full duplex)
- 3-year warranty

Advantages of the digital T35D load cell over the analogue T35 load cell:

- Higher resolution (up to 1,000,000 steps from zero to full scale)
- Intelligent compensation for each load cell
- Easier, faster maintenance with continuous, individual self-diagnosis
- Direct PC connection via RS485 bus and RS232 converter, up to 1200 metres away
- Internal signal filtering
- Higher protection against EMC interference
- Simpler and faster corner adjustment
- Remote diagnostics via modem using a direct PC connection

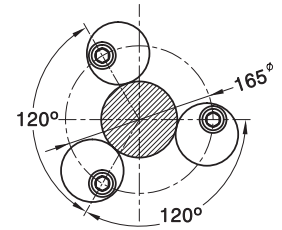
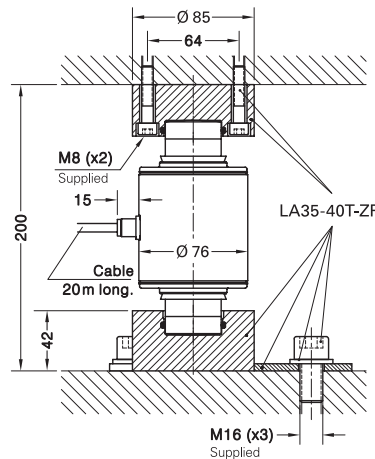
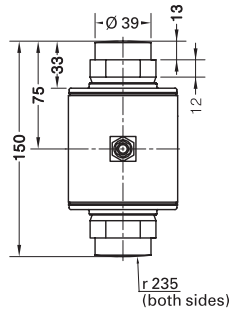
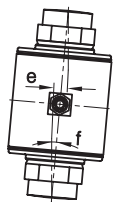


T35D

technical specification...



Mandatory direction of tilt and truck motion



Important: Bumper stops must be fitted to the weighbridge / truck scale and adjusted to limit movement to a recommended min. 3mm, max. 7mm in both the longitudinal and transverse directions. This is essential in order to avoid damage to the equipment. Refer to the critical dimensions e and f below (**).

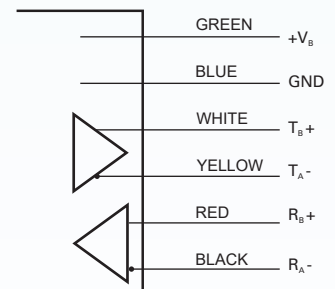
T35D Load Cell

	Load cell Specification		Unit
Load cell Capacities (E _{max})	30, 40		tonnes
Rated Output (S _n)	200000 ± 0.05%		counts*
Accuracy Class	4000	5000	n.OIML
Combined Error	< ± 0.013	< ± 0.01	%S _n
Non-Repeatability	< ± 0.01	< ± 0.008	%S _n
Creep (30 minutes)	< ± 0.012	< ± 0.01	%S _n
Minimum load cell verification interval (V _{min}) E _{max} / Y	E _{max} / 12000	E _{max} / 18000	kg
Temperature Effect on Zero Balance	< ± 0.002	< ± 0.0016	%S _n / °C
Temperature Effect on Span	< ± 0.0012	< ± 0.0010	%S _n / °C
Compensated Temperature Range	-10 to + 40		°C
Operating Temperature Range	-30 to + 70		°C
Safe Overload	200		%E _{max}
Ultimate Overload	350		%E _{max}
Zero Balance	± 1		%S _n
Recommended Supply Voltage	8 - 18		V DC
Maximum Supply Current	60		mA
Conversion Speed	200		Hz
RS-485 serial interface output	Full Duplex		
Maximum transmission cable length	1200		m
Environmental Protection	IP68 & IP69K		
Maximum deflection at E _{max}	0.6 - 1		mm
Cable Length	20		m
Conversion Cable Type	Polyurethane, 6 Core		
Nominal Shipping Weight	Load Cell	3.6	kg
	Accessory	3.7	kg
* User Programmable			

Dimensions

e max.	10.5
f max. permitted tilting	5°

All dimensions are in mm



Electrical Connections

Via 6 wire, 5.7mm diameter, screened polyurethane cable.
Screen not connected electrically to load cell.

Construction

Load cell manufactured from stainless steel

Options

Mounting accessory set comprising nickel plated steel top & bottom mounting cups, centering discs, 3 off M16 bolts and 2 off M8 bolts.
Part number LA35-40T-ZP.

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Our policy is one of continuous product enhancement. We therefore reserve the right to incorporate technical modifications without prior notification.



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